Advances in extracting current profiles from X-band radar images with a focus on retrieving subsurface current

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Abstract

X-band radar images of ocean waves offer a promising approach to estimate shear current near the surface, a region where direct measurement remains challenging. Inversion techniques on wavelength-dependent Doppler shifts have been used previously, however, they focused on the currents deeper in the water column. In our study we concentrate on the upper layer of the ocean closest to the ocean surface. In particular, this study uses a smoothing technique along with in-situ measurements at lower depths to infer the current profile up to the mean sea level, thus delivering a measure for shear current, an important indicator for the air-sea energy transfer. The method was applied to radar images collected during a storm event where concurrent measurements of an ADCP were available. The method also performs well without the use of in-situ measurements at lower depths, though accuracy improves when such data is available to stabilize the solution. The extracted shear magnitude and direction align well with wind measurements taken from 10 m above the sea level providing some validation to our results.

Keywords:
Radar Imaging, Remote Sensing, Oceanography, Ocean Current

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1. Introduction

Processes in the upper few meters of the ocean play a crucial role in governing the exchange of energy, momentum, gases, and other tracers between the ocean and the atmosphere. The dynamics in this region are influenced by a range of factors, including wind stress, wave interactions, buoyancy fluxes, and surface heating or cooling, making it a focal point for understanding air-sea interaction processes (Donelan (1990); Cavaleri et al. (2012)). The upper layer flow also has an influence on the transport of oil spills and positively buoyant pollution.

The vertical profile of the upper layer current can provide critical insight into the distribution of turbulence intensity within this layer. Turbulence, in turn, modulates key physical processes such as the transfer of momentum, heat, and mass across the air-sea interface. For instance, surface turbulence affects the mixing of heat and nutrients into the ocean and the release or uptake of gases like carbon dioxide and oxygen. These processes are pivotal for climate regulation, ecosystem productivity, and the energy balance of the Earth system.

The importance of the upper ocean dynamics is underscored by the identification of two '*Essential Ocean Variables': Surface Current and Ocean Surface Stress*[1]. These variables represent fundamental aspects of the ocean's role in the Earth system, providing standardized metrics to monitor and model the ocean's influence on weather, climate, and marine ecosystems.

In-situ devices like acoustic Doppler current profilers (ADCPs) are incapable of capturing the current just underneath the water surface when deployed on the bottom or a submerged buoy. Measurements from floating buoys also have their complexities as the buoys follow the moving surface providing Lagrangian (when free-floating) or quasi-Lagrangian (when moored) measurement.

Herein, we explore the possibility of using marine X-band radars to infer the current close to the sea surface. The same technique may be adopted by other optical-based remote sensing devices such as video cameras mounted on airborne platforms. The radar backscatter intensity measured in space and time varies with the phase of surface waves and thus contains information concerning wave propagation. As the presence of a background current changes the speeds at which waves propagate compared to quiescent waters, analysis of the radar images can be used to infer the currents and has been widely used in the past as a remote sensing method.

The idea of estimating currents from radar measurements dates back to

[1] https://goosocean.org/what-we-do/framework/essential-ocean-variables/

Stewart and Joy (1974). In their work, Stewart and Joy related the frequency shift in the spectra of a high frequency (HF) radar to the effective current, that distorts the symmetry of the linear dispersion relation of water waves. The linear dispersion relation for water waves with a background current as a function of wavenumber $\vec{k}$ is defined by

$$\omega_{DR}(\vec{k}) = \sqrt{gk \tanh(kh)} + \vec{k} \cdot \vec{U}_{\text{eff}}, \tag{1}$$

where $k = |\vec{k}|$, g is the acceleration due to gravity and $\vec{U}_{\text{eff}}$ is the wavenumber dependent Doppler shift velocity from the background current profile as discussed by Stewart and Joy (1974).

The latter is a depth integrated current where the relevant depth relates to the wavelength of the electromagnetic wavelength during operation. X-band radars typically capture gravity waves in the range of around [15m;200m]. The longer the wave, the more it will be affected by currents in deeper layers, which can be used to infer the current profile. The principal idea of Stewart and Joy (1974) was already successfully applied to generate current maps (Lund et al. (2018)), return average currents (Serafino et al. (2009)), reconstruct near surface currents in deep water ((Lund et al., 2015; Serafino et al., 2012)) and shallow water (Lund et al. (2020)), and invert current profiles (Campana et al. (2016)). None of these studies investigates the current shear just underneath the water surface. The current profiles inverted by Campana et al. (2016) stop some meters below and Lund et al. (2020) only consider the effective current estimated from the distorted dispersion relation. In their conclusion, they state the need for the application of an inversion method.

Inversion of the radar backscatter data to yield the current profile typically involves extracting Doppler shift velocities from the 3D frequency-wavenumber spectrum. Peak spectral energy for a given wavevector is assumed to lie along the wave linear dispersion relation described by Eq. (1). The Doppler shift velocities are wavenumber-dependent when the background flow varies with depth, and extraction over a range of wavenumbers is required to infer information as to the depth profile. The set of wavenumber-Doppler shift pairs are then used as input to an inversion method to find the current depth profile.

A variety of methods have been used to extract the Doppler shifts from the spectrum, among those least-squares (LS)-based methods, polar current shell (PCS) as well as those based on a normalized scalar product (NSP) method. The PCS and NSP method were already applied by Huang et al. (2016). However, they only considered the effective current and did not apply any inversion method. The NSP method was found to perform better than the LS method (Weichert et al. (2023)) in minimizing biases due to spectral leakage, and was used in the present study. Spectral leakage is caused by the window size and observed wavelength not matching which strongly affects low wavenumbers.

Many inversion methods rely on assumptions on the shape or smoothness of the current depth profile (Campana et al., 2016; Lund et al., 2015). The effective depth method (EDM) relies on assuming that the depth profile follows a linear or logarithmic function. A Gaussian quadrature method has been applied to approximate the Stewart and Joy integral (Campana et al., 2016) which requires constraints on the smoothness of the profile. The polynomial effective depth method (PEDM) (Smeltzer et al., 2019) does not require an assumption about the form of the current profile. The PEDM starts with the EDM profile, fits the mapped velocities to a polynomial, and subsequently adjust the coefficients based on theory. The PEDM has been demonstrated to improve the accuracy of the inversion on laboratory data when compared to other methods, though performance on radar data obtained in the field has yet to be demonstrated.

2. Material and Methods

The free surface of the radar image sequence is defined as $I_0(x, y, t)$ where x, y correspond to the horizontal spatial dimensions and t is the time coordinate. The radar images are transformed into the wavenumber-frequency domain using a 3-D fast Fourier transform (FFT) resulting in the power spectrum as

$$P(k_x, k_y, \omega) = |FFT\{I_0(x, y, t)\}|^2, \tag{2}$$

where k_x and k_y are the wavenumber components in the $x-$ and $y-$direction respectively and ω is the angular frequency of the wave components. From the wave dispersion relation in Eq. (1) we can see that only certain triplet combinations of (k_x, k_y, ω) are allowed. Hence, the greatest signal is expected to lie in frequency-wavenumber triplets that satisfy the dispersion relation. The symmetry of the Fourier transform means that there are two wave frequencies $\omega_\pm$ associated with each unique wavenumber: $\omega_+(\vec{k}) = \omega_{DR}(\vec{k})$ and $\omega_-(\vec{k}) = -\omega_{DR}(-\vec{k})$. Below in Fig. 1 the flow of the method sections is illustrated.

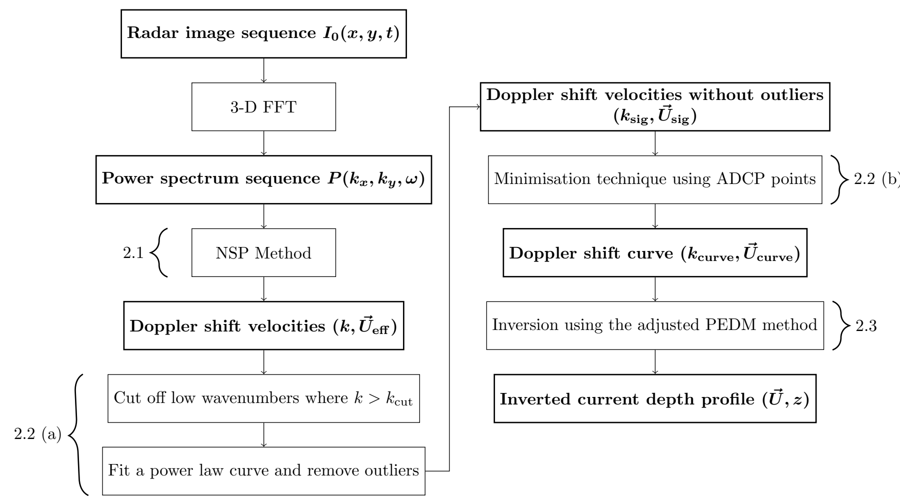


Figure 1: A block diagram showing the method to go from a radar image sequence to the depth profile of the inverted current. The applied methods are labeled as section numbers which correspond to the below sections.

2.1. Normalized Scalar Product (NSP)

In this work a novel adaptation of the NSP method is used as discussed in Smeltzer et al. (2019). This method involves maximising the NSP between the spectrum and a characteristic function G, which defines the dispersion shell, to find the wavenumber-dependent Doppler shift velocity from the wave spectrum. Previous implementations of the NSP have used a characteristic function which is equal to unity within a narrow frequency interval around the dispersion shell and equal to zero elsewhere (Serafino et al. (2009), Huang and Gill (2012), Huang et al. (2016)). Instead, we define the characteristic function G as

$$\begin{aligned} G(\vec{k}, \omega, \vec{U}_{\text{eff}}) = \exp &\left(-2 \left[\frac{\omega - \omega_{\text{DR}}(\vec{k}, \vec{U}_{\text{eff}})}{\delta} \right]^2 \right) \\ &+ \exp \left(-2 \left[\frac{\omega + \omega_{\text{DR}}(\vec{k}, \vec{U}_{\text{eff}})}{\delta} \right]^2 \right), \end{aligned} \tag{3}$$

where $\vec{k} = (k_x, k_y)$, ω_{DR} is defined by Eq. (1) and δ is a parameter that controls the spectral width in frequency of the dispersion relation shell. The terms on the right hand side show the contributions from $\omega_\pm$ respectively.

The normalized scalar product between the spectrum and the characteristic function G is defined as

$$V(\vec{U}_{\text{eff}}) = \frac{\langle |F_i(\vec{k},\omega)|, G(\vec{k},\omega,\vec{U}_{\text{eff}})\rangle}{\sqrt{P_F \cdot P_G}}, \tag{4}$$

where $F_i = \sqrt{P(k_x, k_y, \omega_{DR})}$, P_F and P_G are the respective powers of the image spectra F_i and G respectively. The value of V is maximized with respect to the Doppler shift components, $\vec{U}_{\text{eff}}$. This gives the greatest overlap between the dispersion relation shell and the image spectrum. Hence, we obtain the Doppler shift components, $\vec{U}_{\text{eff}}$, as a function of the wavenumber magnitude $k = |\vec{k}|$.

2.2. Smoothing Method

The Doppler shift velocities are expected to be a smooth quantity as a function of wavenumber. This result is explained further in Section 2.3. In our case, there are some sudden jumps in the NSP points. These jumps can be eliminated through a fitting process applied directly to the NSP points. In (a) and (b), four parameters are introduced: k_{cut}, f_σ, n and z_{cut}.

(a) *Outlier Detection*: The low wavenumber Doppler shifts from the NSP method are strongly affected by spectral leakage as discussed in Weichert et al. (2023). If these Doppler shifts are used, the inversion method will produce high velocities at the lowest depths. Thus, we need to define the wavenumber cut-off point, k_{cut}. The remaining Doppler shifts, ($k > k_{\text{cut}}$), were fitted by a general power law curve. Outliers were detected and removed based on their proximity to the fitted curve, defined as f_σ. We are left with a set of Doppler shifts with outliers removed, $(k_{\text{sig}}, \vec{U}_{\text{sig}})$.

(b) *Minimisation Technique*: We aim to find a realistic curve that fits these Doppler shifts. We construct extrapolations of the ADCP profile up to $z = 0$ m for a range of velocities U_0. The ADCP profile is included up to a maximum depth of z_{cut}. In the interval ($-4 \leq U_0 \leq 4$) m/s, a polynomial of order n is fitted which gives the effective current. Although the range of possible extensions from the ADCP are very different, the effective current only shows minor changes. We can calculate which effective current fits best to the measured Doppler shifts by minimising the root mean squared error (RMSE) between the effective current and the radar Doppler shifts. The corresponding effective current gives the Doppler shift curve $(k_{\mathrm{curve}}, \vec{U}_{\mathrm{curve}})$. This Doppler shift curve will only contain wavenumbers where $k_{\mathrm{curve}} > k_{\mathrm{cut}}$.

The default settings of the parameters are shown below in Table 1 along with the reasoning behind each parameter. A sensitivity analysis of these parameters has been presented in Section 4.1.

Parameter	Value	Reason
k_{cut}	0.11 rad/m	Chosen value based on minimum wavenumber that discards all the very high velocities at lower wavenumbers.
f_σ	σ	Chosen value is a common choice for outlier detection.
n	4	Trade off between avoiding distortion from higher-order models and the simplicity of lower-order models.
z_{cut}	-1.7 m	Includes the full ADCP measurements that were available.

Table 1: Parameter values that have been chosen along with the reasoning behind these choices. Here, σ represents one standard deviation of the residuals.

2.3. PEDM

The depth-dependent Doppler shift velocity can be approximated as a weighted average of the current as a function of depth which was shown by Stewart and Joy (1974) to be

$$\vec{U}_{\mathrm{eff}}(k) = 2k \int_{-\infty}^{0} \vec{U}(z) e^{2kz} dz, \tag{5}$$

in deep water. It is evident in Eq. 5 that inputting any current profile results in a smooth Doppler shift profile. In a finite water depth, h, the Doppler shift velocity was approximated by Skop (1987) and Kirby and Chen (1989) to be

$$\vec{U}_{\mathrm{eff}}(k) = \frac{2k}{\sinh(2kh)} \int_{-h}^{0} \vec{U}(z) \cosh[2k(h+z)] dz. \tag{6}$$

An inversion method is required to find the current profile, $\vec{U}(z)$, using the Doppler shift velocities, $\vec{U}_{\text{eff}}$, found by the NSP method. We use the processed Doppler shift curves that include ADCP information, $\vec{U}_{\text{curve}}$, from Section 2.2 instead of $\vec{U}_{\text{eff}}$. The inversion method is carried out separately on each of the current components. The effective depth method has been extensively used to find estimates of the current profile in Lund et al. (2015) and Fernandez et al. (1996). However, this method requires an assumption of the functional form of the current profile. This can result in errors when the true current profile does not match the assumed form which is common in realistic cases.

The PEDM was proposed by Smeltzer et al. (2019) to prevent a prior assumption of the functional form of the current profile. This method begins with the effective depth method and then fits the profile to a polynomial form. The coefficients are scaled to give an improved estimate of the true current profile. Two small changes were applied to the PEDM from its original use in Smeltzer et al. (2019). In our case the water depth is finite meaning that the initial effective depth is generally defined as

$$Z_{\text{eff}}(k) = -(2k)^{-1} \tanh(kh), \tag{7}$$

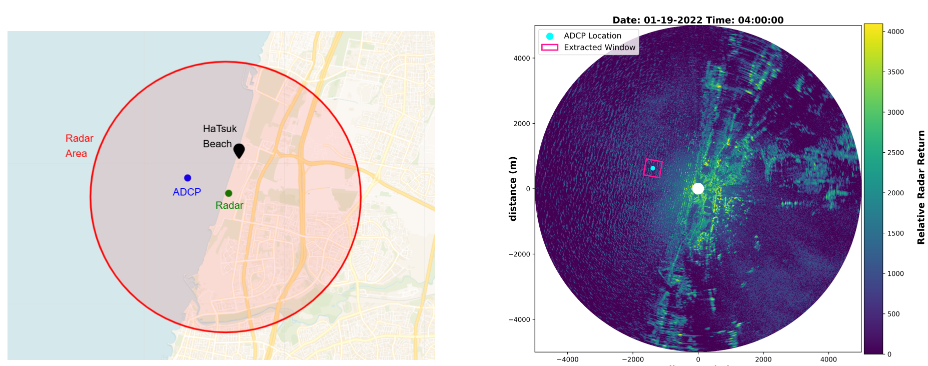
where h is the water depth, approximately 16 m in this study. The remaining steps of the PEDM require no adaptation to account for the finite depth as for most realistic cases the finite depth mapping function yields sufficient accuracy. However, we will add one step to the PEDM to make up for the lack of lower wavenumbers in the Doppler shift curve. We have assumed that we have access to ADCP current measurements in this region. We can use the ADCP current measurements to further refine the PEDM. This is done by adding a step to the PEDM process that is described in Section 2.1.2 of Smeltzer et al. (2019). This passage outlines that "step 5" has created a set of linearly extrapolated points down to z_c based on the average shear. Firstly, we find the ADCP depth that is closest to the surface which we label z_a. We look for the extrapolated point that is just beneath the value of z_a. If this value exists we use the combination between the ADCP measurements below z_a and the extrapolated points above z_a. Then we perform the final polynomial fit as outlined in "step 6". This allows us to specify the velocity profile at discrete depths within the final PEDM fitting.

3. Data overview

3.1. X-band radar data

X-band radar-based oceanographic measurements were carried out using a WaMoS II system on a fixed installation. The marine radar station was located at $(32^\circ\ 07'\ 50.0''N,\ 34^\circ\ 47'\ 14.0''E)$ near HaTsuk beach. It captured radar images of a circular region with a radius of 4 km with the radar station at its centre, including a section of ocean and a section of land. The radar operated with a rotation period of 2.46 seconds with a range resolution of 3.75 m. For a good frequency resolution, radar sequences of 15 min were analysed as discussed in Støle-Hentschel et al. (2023). The radar sequence contained 366 images. Radar sequences of 15 min were extracted every hour. We studied a storm with high wind speeds and significant wave heights starting at 22:00, 18$^{\text{th}}$ January 2022 until 00:00, 20$^{\text{th}}$ January 2022. In such conditions, it is assumed that a strong shear can develop.

The ADCP was located at $(32^\circ\ 08'\ 04.8''N,\ 34^\circ\ 46'\ 27.7''E)$ which was ~ 1.5 km off the coast as shown in Fig. 2. We took a 500×500 m extraction window meaning that each radar image had 132×132 pixels. We positioned the extraction window with the ADCP at its centre so that we could compare the ADCP measurements to the radar measurements. The extraction window was rotated clockwise 80° so that the extracted window was in the main wave direction.



(a) Map

(b) Radar Image

Figure 2: Visual representation of the area mapped by the radar. (a) Area captured by the radar image north of Tel-Aviv. (b) The first $t = 0$ radar image of the sequence of images.

3.2. ADCP processing

A Nortek's Signature 1000 ADCP was deployed up-looking mounted on the sea bed at water depth of $\sim$ 16 m offshore Tel Aviv, Israel, in front of the X-band Radar. A vertical acoustic surface tracking beam measured the sea elevation, and four slanted beams measured the particle scatter from the water column, indicating the east and north current velocities, at sampling frequency of 4 Hz. The raw ADCP data was extracted and was corrected with magnetic declination of 4.9°, where bursts of 4096 samples (corresponding to $\sim$ 17 min) were processed for each time. Side lobe effects were eliminated from the top 10% height level, and correlations of less than 50% were removed. A minimum amplitude level was set as 35dB, and in case of time series at some depth cell including less than 80% quality data, the complete depth level data was removed.

3.3. Environmental conditions

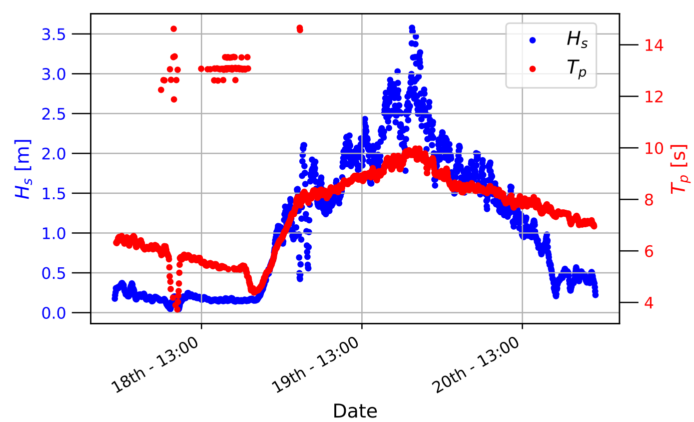


Figure 3: The times series of the mean significant wave height (H_s) and mean peak period (T_p) from 18 Jan 2022 00:00 to 21 Jan 2022 00:00. The values are calculated by WaMoS II for the window in the main wave direction (around the ADCP) and averaged over 20 minutes.

We studied the development of a strong winter storm that occurred during the period from 22:00, January 18, 2022 to 00:00, January 20, 2022. According to estimates available from the WaMoS II system of the X-band radar, the 20-min average of significant wave height (H_s) increased from 0.2 to 3.6 m within 24 hours (Fig. 3), with the maximum significant wave height reaching 4.8 m (not shown). The mean peak period varied between 4.5 and 10.2 s (Fig. 3), with some larger values around 13 s attributed to a swell. Measurements of H_s by an ADCP within the same observation window deviate less than 0.5 m, most likely due to the sampling variability in the different spatial coverage of the measurements.

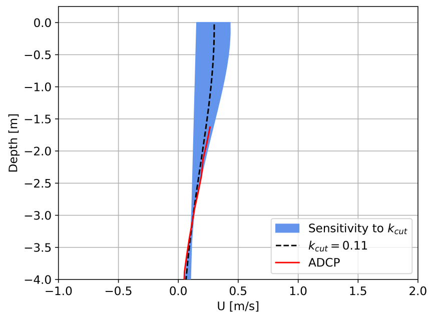
For accurate wave and surface current measurements, WaMoS II needs a minimum wind speed of 3 m/s and wave height larger than 0.5 – 0.75 m. According to the Meteorological Service (IMS, `https://ims.gov.il/en/data_gov`), the wind speed varied from 4 to 15 m/s for the time period of interest at the nearest Tel-Aviv coastal station. Thus, both wind and wave characteristics satisfy the above requirements, allowing for a reliable analysis of X-band data and current inversions.

4. Results

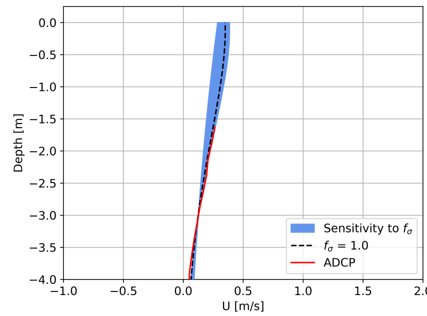
4.1. Parameter Sensitivity

The wavenumber-dependent Doppler shifts were calculated for each time period using the NSP method followed by the smoothing method outlined in Section 2.2. As discussed, four parameters were required for the application of the method. To ensure that the method is robust for future use, we conducted a sensitivity analysis of these parameters on the resulting Doppler shift curve. The parameters are k_{cut}, f_σ, n and z_{cut}. To investigate the sensitivity of each parameter, one parameter will be varied while the other parameters are set as the default settings that are shown in Table 1. We will test the parameters using the East velocity of the 19 Jan 2022 11:00 case.

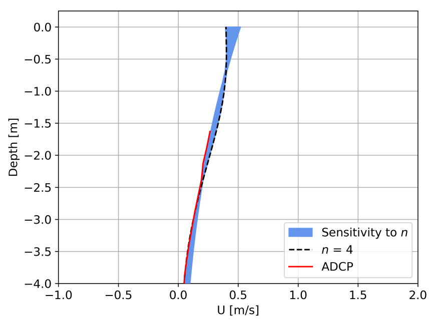
In Fig. 4, the results of the sensitivity analysis are shown. The parameter k_{cut} was varied between 0.01 rad/m and 0.2 rad/m. The parameter f_σ was varied between 1 and 7 inclusive. The parameter n was varied between 0.6 and 3 standard deviations. The parameter z_{cut} was varied by setting the values as $z_{\mathrm{cut}} = -3, -6, -9, -12$ m. The blue area represents the area that is covered by all the different parameters tested.



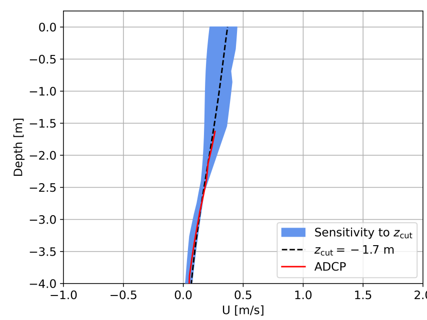
(a) k_{cut}



(b) f_σ



(c) n



(d) z_{cut}

Figure 4: Shows the sensitivity of the inverted currents to the parameters (a) k_{cut}, (b) f_σ,(c) n and (d) z_{cut}. The blue area shows the current inversions for all of the various parameters studied.

It is evident in Fig. 4 that the final inverted current is sensitive to k_{cut}. The areas for both f_σ and n are very small reflecting that they have a very small impact on the final inverted current. The sensitivity to k_{cut} is intuitive as it chooses where the lower wavenumbers are cut-off which is the region that spectral leakage is particularly impactful. For all time steps, the average k_{cut} value is very close to 0.11 rad/m making this a reasonable choice. The inverted currents are also sensitive to z_{cut} which chooses how much of the ADCP measurements are used in the inversion process. Generally, the most extreme current gradient occurs when z_{cut} is deeper in the water column and is milder towards $z = 0$ m.

4.2. Inverted currents

The current depth profile is found by using the PEDM to invert the Doppler shift measurements. The PEDM can be performed on the raw Doppler shift velocities $(k, \vec{U}_{\text{eff}})$ however this causes unrealistically high velocities at lower depth. To constrain the inverted current, we have employed an outlier detection and minimisation technique as outlined in Section 2.2. This uses a combination of radar and ADCP measurements and define the Doppler shift curve $(k_{\text{curve}}, \vec{U}_{\text{curve}})$. Wavenumbers below the cut off value k_{cut} are not included in the inversion. The PEDM can be used on this smooth Doppler shift curve which is referred to as the standard PEDM.

The results obtained using the standard PEDM are reasonable as shown for the 32 current profiles in Fig. 5. However, for a number of time steps, the standard PEDM struggles to invert the current. In these instances, the wind and waves are not aligned. Thus, it seems to be necessary for the wind and wave directions to be aligned for the standard PEDM to obtain reasonable currents.

By inspecting Eq. (5), it is evident that smaller wavenumbers are influenced by currents at greater depths. There are no wavenumbers which are less than $k_{cut} = 0.11$ rad/m in the Doppler shift curve. To account for this an adjusted PEDM is proposed in which the current profile is specified at discrete ADCP depths. The adjusted PEDM can be used on the Doppler shift curve which results in the inverted current shown in red in Fig. 5. In contrast to the standard PEDM, this enhanced version delivers reliable current inversions in all cases. The RMSE is calculated for each time step for both the standard PEDM and adjusted PEDM against the ADCP between $z = -6$ m and the uppermost ADCP measurement. The results of this are shown in Table 2, in which it is evident that the adjusted PEDM performs better than the standard PEDM. The adjusted PEDM is discussed further in Section 4.3.

	Standard PEDM	Adjusted PEDM
Average RMSE	0.076	0.013
Maximum RMSE	0.368	0.053
Minimum RMSE	0.010	0.003

Table 2: Descriptive Statistics for the RMSE for both the standard PEDM and Adjusted PEDM when being compared with the ADCP respectively.

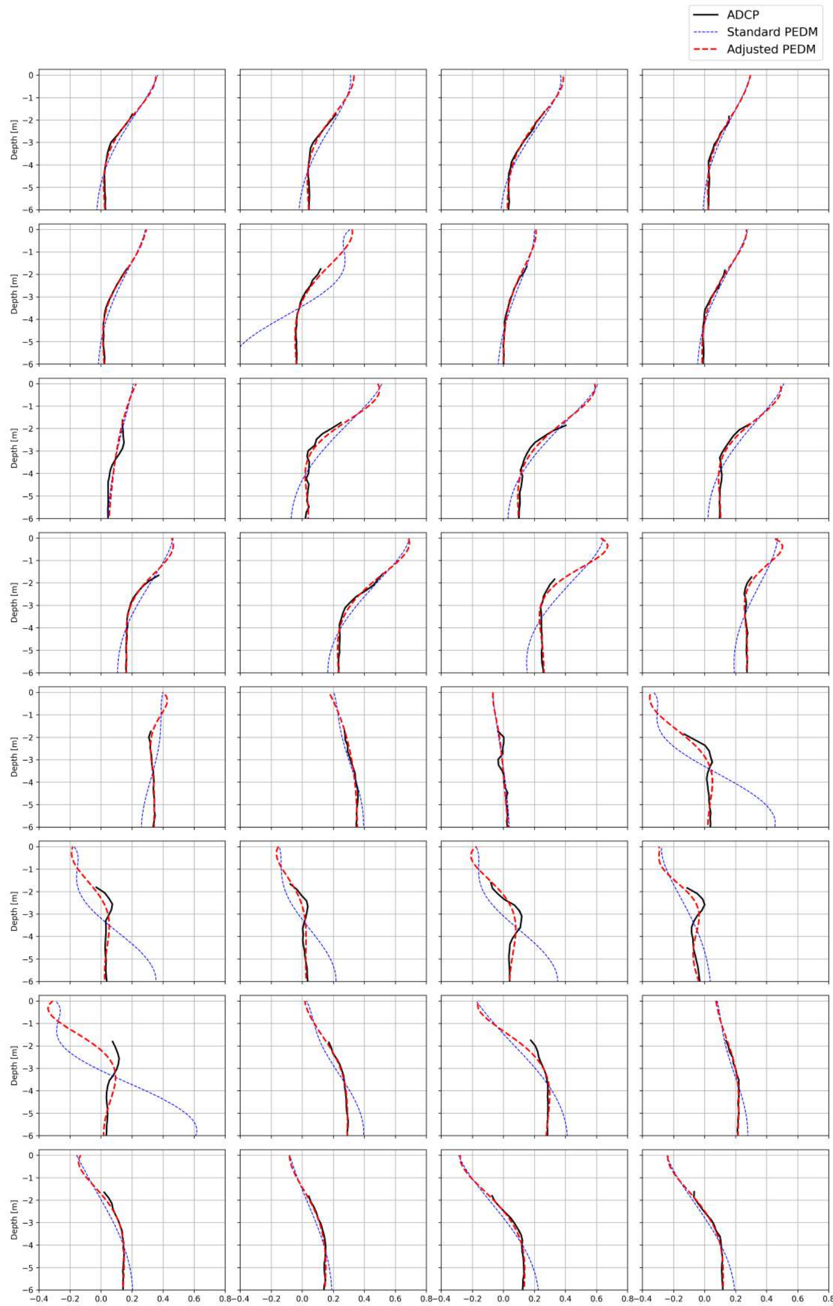
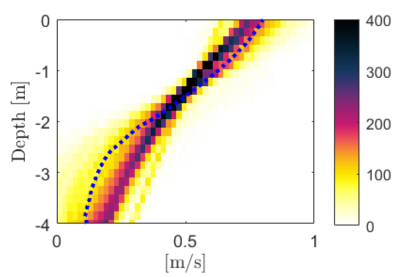


Figure 5: Current depth profile in the uppermost layer of the water column in the North and East directions for all 16 time periods. The inversion is performed using the adjusted PEDM shown in red and the standard PEDM shown in blue.

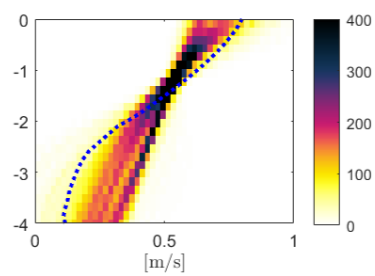
It is important to note that the method performs reasonably well for both the standard PEDM and adjusted PEDM. As will be shown in Section 4.4, the resulting shear in the uppermost layer and the gradient in the ADCP profile underneath are not statistically dependent, despite the inclusion of the ADCP information in the current inversion.

4.3. Doppler shift uncertainty propagation: inclusion of ADCP velocity-depth pairs

The inclusion of velocity-depth pairs from ADCP data in the PEDM algorithm has been shown in Sections 4.2 to improve the inversion process for shallower depths. To study in greater detail the improvements from including ADCP data, mock Doppler shifts were generated with the addition of many realizations of random noise, and inverted to produce depth profiles with and without ADCP depth-velocity pairs for each realization. As an example, the North current velocities from case 18 Jan 2022 23:00 were used as the true current profile, extrapolating to the surface by adding a depth-velocity pair [0 m, 0.8 m/s] to the existing dataset, and thereafter performing a spline interpolation. Doppler shift velocities were evaluated using (6) over a representative wavenumber range spanning 0.075 to 0.34 rad/m. Noise with standard deviation π/Tk was added to the computed Doppler shift velocities, with $T = 10$ min (a representative duration of X-band dataset). The noise thus had a standard deviation in frequency equal to half the assumed spectral resolution, with an assumed $1/k$-propagation to the Doppler shift velocities.



(a) With



(b) Without

Figure 6: 2D histograms of PEDM current inversion with (a) and without (b) the inclusion of exact depth-velocity pairs up to 4 m depth. The colorbar shows total counts out of 1000 total realizations. The dashed red curve is the exact current profile.

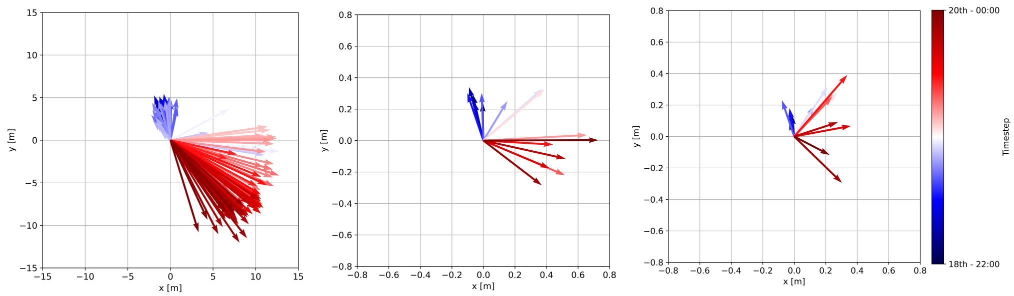
The Doppler shift velocities were then used as input to the PEDM, with and without inclusion of the exact depth-velocity pairs up to 4 m depth. The process was repeated over 1000 realizations. Fig. 6 shows a 2D histogram of the resulting current profiles with and without inclusion of the pairs in panel (a) and (b), respectively.

As can be seen, the inclusion of the constraining depth-velocity pairs greatly reduces the spread in the resulting inverted depth profiles. It also improves the accuracy with respect to the true profile (shown as the dotted blue curve). The effect is greatest at deeper depths as expected; the PEDM becomes less accurate at deeper depths as the waves are fundamentally less-sensitive to currents at greater depths.

4.4. Validation using wind measurements

In the absence of ground truth, the current profiles were validated by comparing them to wind which is considered a reliable geophysical measurement. To simplify the comparison to wind, each current profile was represented by its mean gradient. The wind magnitude and direction are expected tp have a similar evolving pattern over time as the gradient of the upper layer current. We have found that this holds for the gradient of our calculated current profile. In particular, during the period studied there was a notable change in wind speeds and direction at around 19 Jan 2022 09:00 which is reflected in the inverted currents.

The wind data comes from an open source dataset on the Meteorological Service (IMS) database. The nearest IMS station, named 'Tel Aviv Coast', is located at ($32^\circ\ 03'\ 28.8''N$, $34^\circ\ 45'\ 31.7''E$) which is deemed sufficiently close to the radar location for comparison. The wind speeds and directions represent a scalar average over the 10 minute period measured at 10 m. The distinct change of directions in Fig. 7a can also be observed in the directions of the currents. There is an agreement in the general evolution of the vectors of the wind in Fig. 7a and the currents in Fig. 7b and c.

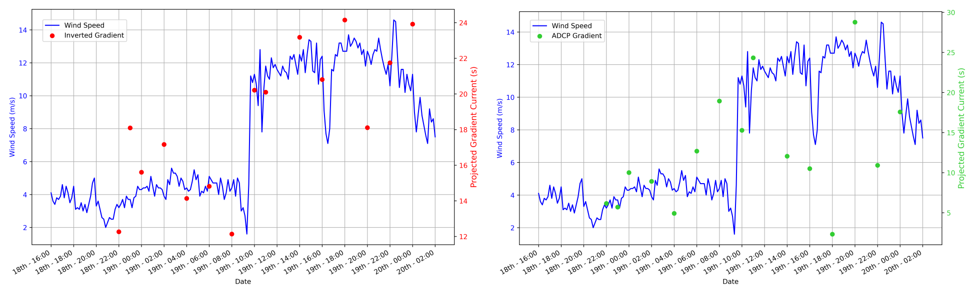


(a) Wind Vectors (b) Inverted Current (c) ADCP Current

Figure 7: Time evolution of the (a) wind vectors from the IMS station, (b) inverted current at $z = 0$ m and (c) ADCP current at $z = -1.7$ m. Dark blue represents the start of the studied time frame 18 Jan 2022 22:00 and dark red represents the end of the studied time frame 20 Jan 2022 00:00.

It can be observed in Fig. 7 that the direction of the radar current is more aligned with the wind after the wind increase than is the case for the ADCP. This is expected due to the rotation of the direction downwards in the water column.

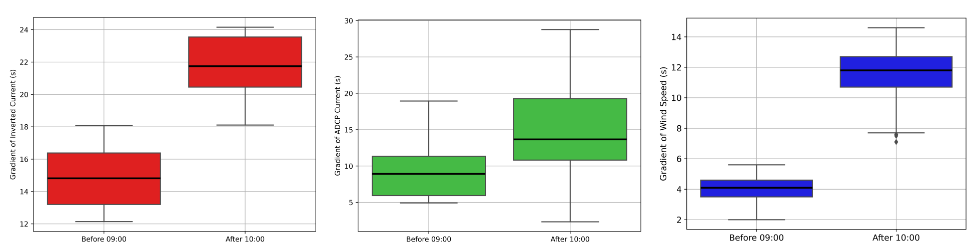
For comparison with the wind, the magnitude of the current was calculated from the North and East component before it was projected to provide the component along the wind direction. The gradient of the current profile derived from the radar was calculated as the mean gradient between $z = 0$ m and $z = -1.7$ m. whereas the shear of the ADCP was derived from the gradient of the ADCP profile between $z = -1.7$ m and $z = -3$ m. The upper bound is the highest depth that the ADCP is available to. The difference the upper bound between the definition of shear for the inverted current and the ADCP does not have an impact on the evolution of the values, only on the magnitudes themselves. The current gradients are shown below in Fig. 8. The gradient value for 19 Jan 2022 12:00 is dismissed as it is an order of magnitude larger than the other gradient values. This large value could be a late response to the observed wind speed increase in this region.



(a) Projected Inverted Current Gradient (b) Projected ADCP Current Gradient

Figure 8: Times series of (a) the projected inverted current gradient and (b) the projected ADCP current gradient over the studied time period. They are plotted on different scales to help with visual comparison.

The large increase in wind speed occurring between 09:00 and 10:00 on January 19$^{\text{th}}$, 2022 (compare Fig. 8) caused a statistically significant difference in the median for the for the inverted currents but not for the and ADCP currents. This could be shown using a Mann-Whitney U test, a non-parametric test that can be applied to a small sample size and that does not require the statistics to follow normal distributions before and after the wind increase. As required by the test, the before and after groups are statistically dependent. The Mann-Whitney U test was performed separately for the inverted current gradients, the ADCP current gradients and the wind speed with the respective boxplots shown in Fig. 9.



(a) Inverted Gradient (b) ADCP Gradient (c) Wind Speed

Figure 9: Box plots showing the difference between before 09:00 and after 10:00 for (a) inverted current gradient, (b) ADCP current gradient and (c) wind speed in m/s.

The calculated p-values are $p \approx 0.001$ for the inverted current gradient, $p \approx 0.15$ for the ADCP current gradient and $p \approx 0$ for the wind speed. A common choice of the significance level is $\alpha = 0.05$. Hence, we can conclude for both the wind speed and the gradient current that there is a statistically significant difference between the medians before 09:00 and after 10:00 on January 19$^{\text{th}}$, 2022. However, for the ADCP current gradient the p-value is larger than the significance level meaning that we cannot conclude that there is a statistical difference in the medians before 09:00 and after 10:00 on January 19$^{\text{th}}$, 2022. We have found that both the wind matches the trend over time of the gradient of our calculated current profile, however the ADCP does not capture this behaviour suggesting that our method has captured relevant information in the upper layer.

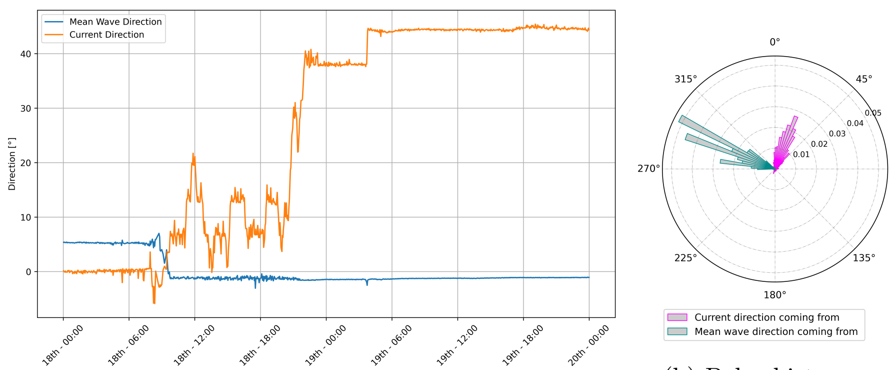
5. Discussion

5.1. Influence of the window size

It is well known that long waves require large windows to avoid spectral leakage. The initial choice of the window size of square boxes with a length of 500 m was chosen as a trade-off between capturing intermediate to short wind waves while observing an area of homogeneous conditions. A test of doubling the side length of the window did not improve the estimates of longer waves.

5.2. Stokes drift is negligible

Two-dimensional wave spectra provided by WAMOS were used to calculate the Stokes drift. The values were at least one order of magnitude lower than the Doppler shifts. Therefore, their impact was not considered separately.



(a) Times series of unwrapped directions.

(b) Polar histogram

Figure 10: Times series and histogram of the current direction and mean wave direction from 18 Jan 2022 00:00 to 20 Jan 2022 00:00. Polar histogram is based on 974 observations and a bin width of 4°.

In Fig. 9 it is evident that the radar inversion experiences a larger variation in current gradient for slower wind. However, when the wind speed increases we observe that the variation for the radar inversion gets much smaller. This could be attributed to the improved radar image when winds are stronger.

Another possible explanation is the influence of the directional spreading of the waves in combination with the relative direction between waves and current (see Fig. 10). The two directional wave spectra obtained by the WAMOS system give insight in the directional spreading of wave energy. Initially, wave energy is distributed over 90°, which allows an omnidirectional estimation of currents. When the wind increases and changes direction, the wave energy is concentrated around the peak direction. This results in good current estimates aligned to the main direction. However, current components perpendicular to that direction cannot be captured. Fig. 10 shows that the relative angle between current waves is about 45° during the second half of the observation period, which can affect the results due to the low directional spreading of wave energy. This could also in part explain the large errors of the estimated Doppler shifts of longer waves. The directional spreading of those waves was lower and thereby impedes the success of the method.

5.4. Limitations of the method

In Section 4.1 the dependence on the parameters was discussed. The inverted currents were sensitive to the amount of ADCP measurements that were used to constrain the method (z_{cut}) at the lower depths. The method has inherent dependence on these ADCP measurements, however it has been shown that for some cases less ADCP measurements can also provide reasonable results. In particular, cases in which the wind and the waves were aligned. More work would be needed to better study the errors within the method to classify which cases would give reasonable upper layer currents whilst using less ADCP measurements.

6. Conclusion

In the present work, the current profile is inferred from X-band radar images up to the mean sea level. It could be shown that a sudden increase of the wind speed lead to a statistically significant increase in the median upper layer shear measured by the radar. The novel approach relies on a smoothing technique that incorporates in situ measurements from an ADCP at lower depths to infer the current profile. Although the increase in shear could also be observed in the ADCP data (that does not include the uppermost layer), the response to the storm was less pronounced and not statistically significant.

The presented results are a proof of concept that are planned to be investigated with more exhaustive data in a follow-up study. Future development of the method should tackle to obtain independence of an in-situ sensor, either by filtering errorneous cases or by replacing the information at lower depths from global current models. Remote sensing could thus be used to enhance global models with predictions of upper layer shear.

CRediT authorship contribution statement

Joseph Anderson: Writing – review & editing, Writing – original draft, Visualization, Data curation, Formal analysis, Methodology, Software. **Benjamin K. Smeltzer:** Writing – review & editing, Visualization, Data curation, Software. **Rotem Soffer:** Writing – review & editing, Data curation. **Yaron Toledo:** Writing – review & editing, Conceptualization, Data curation, Validation, Methodology. **Vika Grigorieva:** Writing – review & editing, Visualization. **Frédéric Dias:** Writing – review & editing, Conceptualization, Data curation, Validation, Supervision. **Susanne Støle-Hentschel:** Writing – review & editing, Visualization, Conceptualization, Data curation, Validation, Supervision, Resources, Methodology, Software, Project administration.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

Data available from the corresponding author upon reasonable request.

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